

# Data Flow Diagram

|               |                                                      |
|---------------|------------------------------------------------------|
| Date          | 07 July 2026                                         |
| Project Name  | CaptionAI: AI Powered Social Media Caption Generator |
| Maximum Marks | 2 Marks                                              |

## Purpose

This Data Flow Diagram (DFD) illustrates how data moves through the CaptionAI system. It shows how the user's input is processed, how the AI model generates the caption, how the generated content is stored, and how the final results are delivered back to the user.

## DFD Symbol Legend

| Symbol                            | Name            | Description                                                                                                  |
|-----------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------|
| Oval / Rounded shape              | External Entity | A person, organization, or system outside the project that sends or receives data (e.g., User, Groq API).    |
| Rectangle with numbered header    | Process         | An activity that transforms incoming data into outgoing data (e.g., Caption Generation, History Management). |
| Rectangle (solid fill, no number) | Data Store      | A place where data is held for later use (e.g., User Database, Caption History Database).                    |
| Labeled arrow                     | Data Flow       | The movement of data between entities, processes, and data stores, labeled with what data is being passed.   |

## External Entities

**User** – Enters the content prompt, selects the platform and preferred tone, views the generated caption, and can save, copy, or download the results.

**Groq API (LLaMA 3.3-70B Versatile Model)** – The external AI service that receives the caption generation request and returns the generated caption, hashtags, and call-to-action suggestions.

## Processes

**1 – User Authentication** – Verifies the user's login credentials and provides secure access to the application.

**2 – Caption Generation** – Accepts the user's prompt, selected platform, and tone, sends the request to the Groq API, and receives the generated caption along with the hashtags and call-to-action suggestions.

**3 – History Management** – Stores the generated captions in the Caption History Database and retrieves previously generated captions when requested by the user.

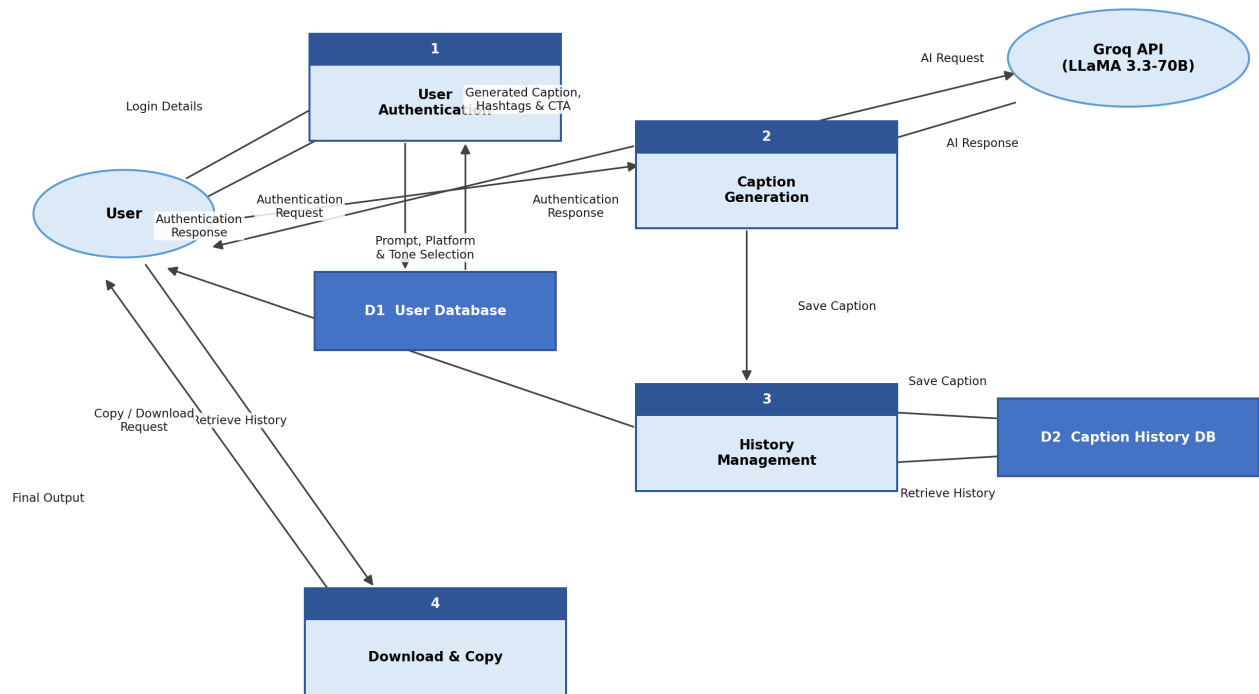
**4 – Download & Copy** – Allows the user to copy or download the generated caption for future use.

## **Data Stores**

**D1 – User Database** – Stores user account details, login credentials, and profile information.

**D2 – Caption History Database** – Stores generated captions, hashtags, CTA suggestions, prompts, selected platform, selected tone, and the generation date and time.

## DFD Diagram



## Process Flow

- 1 The user logs into the application.
- 2 The system verifies the user's credentials using the User Database.
- 3 After successful login, the user enters a prompt and selects the platform and preferred tone.
- 4 The application sends the request to the Groq API (LLaMA 3.3-70B Versatile Model).
- 5 The AI model generates a caption, hashtags, and call-to-action suggestions.
- 6 The generated content is stored in the Caption History Database.
- 7 The generated caption is displayed to the user.
- 8 The user can copy, download, or view previously generated captions from the history section.